

- Sanger, Frederick, 416
 Sapwood, 774f
 SAR clade, 599f, **601f**–608f, 609
 Sarcomeres, 879f, **1126f**
 Sarcoplasmic reticulum (SR), **1128f**–1129f
 Sarcophagii (lobe-fins), 728, 729f, 730
 Sargasso Sea, 1242, 1243t
 Sarin, 158, 1080
 Satellites, determining primary production with, 1242f
 Satiety center, 917f
 Saturated enzymes, 156
 Saturated fats, 73f, 74, 128f
 Saturated fatty acids, **73f**, 74
 Savannas, 750, **1174f**
 Savory tastes, 1123f–1124f
 Sawfish, 1020
 Scaffolding proteins, **228f**, 229
Scala naturae (scale of nature), Aristotle's, 470
 Scale bar in Scientific Skills Exercise, 99
 Scale-eating fish (*Perissodus microlepis*), 500, 501f
 Scales
 fish, 728
 reptile, 735
 Scallops, 701f
 Scanning electron microscope (SEM), **94**, 95f, 96
 Scanning electron microscopy (SEM), 95f
 Scarlet fever, 403
 Scatter plots in Scientific Skills Exercises, 54, 136, 205, 513, 751, 1217
Sceloporus species, climate change and, 11f
Schindleria brevipinguis, 719
Schistosoma mansoni, 696f
 Schistosomiasis, 696f
 Schizophrenia, 372f, **1102f**
 Schmidt-Nielsen, Knut, 1136
 Schwann cells, **1076f**–1077f, 1090f
 Science, **16**
 biology as study of life by, 3. *See also* Biology
 cooperative approach and model organisms in, 22–23, 24f. *See also* Model organisms
 diverse viewpoints in, 24
 inquiry process of, 17, 18f–21f, 22. *See also* Case studies; Inquiry Figures
 medical. *See* Medicine
 methods. *See* Research Method Figures
 skills. *See* Scientific Skills Exercises
 society, technology, and, 23, 24f
 Scientific inquiry, 16–20. *See also* Case studies;
 Inquiry Figures; Research Method Figures;
 Scientific Skills Exercises
 Scientific method, 17, 18f–21f, 22. *See also* Case studies; Inquiry Figures; Research Method Figures; Scientific Skills Exercises
 Scientific notation in Scientific Skills Exercise, 1082
 Scientific Skills Exercises. *See list*, xxi
 Scion, **835**–836
 Sclera, 1118f
 Sclereids, 764f
 Sclerenchyma cells, **764f**, 770
 Sclerenchyma tissue, 868f
 Scolex, 697f
 Scorpions, 707, 708f
 Scrapie, 412
Scr gene, 545
 Scrotum, **1025f**
 Scrub jays, 1099, 1162
 Scutellum, 829f
 Scyphozoans, 687f, 691f–692f
 Sea anemones, 692f–693f
 Seabirds, 1238f, 1244
 Sea cucumbers, 715f
 Sea daisies, 713, 714f
 Sea horse, 729f
 Sea lampreys, 723f, 724
 Sea lettuce, 610f
 Sea lilies, 715
 Seals, 44f, 874f, 949f
 Sea otters, 1189
 Sea slugs, 700f, 1020f
 Seasonality, 1167f
 Seasonal turnover, lake, 1178f
 Sea spiders, 707–708
 Sea squirts, 721f
 Sea stars, 689f, 713, 714f, 898f, 940f, 1086f, 1192f, 1226f
 Sea urchin (*Centrostephanus rodgersii*), 1281f
 Sea urchins, 509f, 689f, 713, 715f, 1044f–1047f, 1184f–1185f, 1189
 Sea wasps, 692f
 Seawater, 977f, 979f, 982f
 Seaweed, 599f, 602, 603f, 604, 609f, 610, 1184f
 Secondary cell walls, **118**
 Secondary consumers, **1240f**
 trophic efficiency of, 1246, 1247f, 1248
 Secondary endosymbiosis, **596f**–597f
 Secondary growth, plant, **766**, 783
 cork cambium and periderm production in, 774
 evolution of, 774
 meristem generation of cells for, 766f–767f
 vascular cambium and secondary vascular tissue for, 772f–775f
 Secondary immune response, **962**, **963f**
 Secondary oocytes, **1029f**
 Secondary production, **1246**
 in arctic tundra ecosystem, 1256f–1257f
 production efficiency in, 1246f
 in salt marsh ecosystem, 1247
 trophic efficiency and ecological pyramids in, 1246, 1247f, 1248
 Secondary structure, protein, **80f**
 Secondary succession, 1230–**1231**
 Secondary vascular tissue, 772f–774f
 Second law of thermodynamics, 143, 145f, **146**, 1239
 Second messengers, **223**, **844**
 calcium ions, inositol trisphosphate, and diacylglycerol as, 224, 225f
 cyclic AMP as, 223f–224f
 in hormone pathways, 1002, 1003f
 in plant signal transduction, 844f, 845
 Secretin, 915f, 1005f
 Secretion, excretory system, **984f**
 Secretions
 animal hormone, 1001f–1004f
 cell signaling and. *See* Animal hormones; Neurohormones
 digestive, 898, 907f–908f, 909–910, 915f
 protective, 898
 Secretory phase, 1032f, 1033
 Secretory proteins, 105
 Secretory systems, prokaryotic, 577
 Secretory tubules, marine bird, 982f
 Sedentarians, 703f–704f, 705
 Sedimentary strata, 470f, 529f
 Seed coat, **828f**–829f
 Seedless vascular plants, **622**
 gametophyte-sporophyte relationships in, 637f, 638
 importance of, 633f, 634
 life cycles of, 630f
 origin and traits of, 628f–631f
 phylogeny of, 622f, 631, 632f, 633
 Seedling development, 829, 830f
 Seed plants. *See also* Angiosperms; Gymnosperms;
 Plants
 adaptations of, 653
 angiosperms, 637f, 638, 644f–650f. *See also* Angiosperms
 gymnosperms, 637f–643f
 importance of, to Earth's ecosystems, 636f
 importance of, to human welfare, 651f, 652
 phylogeny of, 623f
 seeds and pollen grains as terrestrial adaptations of, 637f–638f, 639
 threats to biodiversity of, 651f, 652
 Seeds, **622**–623, 636–**637**, **828**
 abscisic acid in dormancy of, 852
 development of, 826, 827f–829f
 dispersal of, 832f
 dormancy of, 639
 evolutionary advantage of, 639
 fruits and angiosperm, 645f
 gibberellins in germination of, 851f
 glyoxysomes in, 112
 phytochromes in germination of, 856, 857f
 plant seedlings development from, 828f–830f
 strigolactones in germination of, 855
 Segmentation, 1066
 Segmented worms, 688f
 Segregation, law of, 271f–275f, 296, 297f
 Seizures, 1076
 Selective breeding, 474, 475f, 841
 Selective permeability, **131**, 132f, 1071f
 Self-assembly, protocell, 527, 528f
 Self-assembly stage, phage lytic cycle, 402f
 Self-fertilization, 834, 835f, 840–841, 1042
 Self-incompatibility, **834**, 835f, 840–841
 Selfing, 834, 835f, 840–841
 Self-pollination, 271f
 Self-pruning, 786
 Self-replicating molecules, 526, 527f–528f
 Self-splicing, 346
 Self-thinning, 801
 Self-tolerance, 961
 Semelparity, **1200**, 1201f
 Semen, **1026**, 1034
 Semicircular canals, **1113f**, 1115f, 1116
 Semiconservative model, DNA replication, **321f**–322f
 Semilunar valves, **927f**
 Seminal vesicles, 1025f, **1026**
 Semiferous tubules, **1025f**
 Senescence, **853**–854
 Senile dementia, 83
 Sensitive period, **1144**, 1145f
 Sensitive plants (*Mimosa pudica*), 802, 862f
 Sensors, homeostatic, **882f**
 Sensory adaptation, **1109**–1110
 Sensory areas, cerebral cortex, 1096, 1097f–1098f
 Sensory input, 1068f, 1096, 1097f–1098f
 Sensory neurons, 1068, 1069f, **1088f**
 Sensory pathways, 1108f–1109f
 Sensory reception, **1108f**, 1109
 Sensory receptors, **1108f**–1111f, 1112
 Sensory systems
 animal and plant, 894f
 arthropod, 707f
 mechanoreceptors for hearing and equilibrium in, 1112f–1116f
 motor systems and, 1107f–1108f. *See also* Motor systems
 muscle function and, 1125, 1126f–1131f, 1132
 sensory receptors in, 1108f–1111f, 1112
 shark, 727
 skeletal systems, locomotion, and, 1132f–1135f, 1136
 snake, 738
 taste and smell receptors in, 1123f–1125f
 visual receptors in, 1117f–1122f, 1123
 Sensory transduction, **1109**, 1120f
 Sepals, **644f**, **823f**
 Separase, 240
 Separate electron orbitals model, 35f
 Septa, **656f**
 September equinox, 1167f
 Septic shock, 957, 976
 Sequence-based phylogenetic tree in Scientific Skills Exercise, 411
 Sequence logo in Scientific Skills Exercise, 351
 Sequencing by synthesis, 415f–417f. *See also* DNA sequencing
Sequoiadendron giganteum, 774f
 Sequoia trees, 643f, 774f
 Serial endosymbiosis, **534**, 535f
 Serine, 77f, 222
 Serotonin, 1081t
 Serpentine plant communities, 30f
 Sertoli cells, 1031f
 Serum, 934
 Seta, **625**
 Set point, homeostatic, **882f**
 Severe combined immunodeficiency (SCID), 434f, 971
 Sewage treatment, 586, 591
 Sex
 chromosomal basis of, 298f, 299
 genomic imprinting and, 310f, 311
 Sex attractant, 43
 Sex chromosomes, **257**
 aneuploidy of, 309
 human, 256f–257f, 258
 patterns of inheritance of, 298f–300f
 Sex determination, 1030
 Sex hormones
 endocrine disruptors and, 1015
 as environmental toxins, 1276
 functional groups and, 62f
 in regulation of mammalian reproduction, 1030, 1031f–1032f, 1033–1034
 in sex determination, 1030
 smooth ER synthesis of, 104–105
 as steroids, 75f, 1014, 1015f